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Last Change	Review	Approval

Document History

Version	Date	Changes	JIRA-Ticket	Name
V1	2025-10-28	Release of v1.0.0 of MR-Bloch-Simulator	n/a	nmo

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1 Introduction and Overview

This document summarizes the features of the MR-Bloch-Simulator. The simulator allows the visualization of the magnetization evolution of a single atom under the influences of radio frequency pulses. Additionally, a multi-atom view is presented. This view includes the simulation of slice selection through the application of a z-gradient field and RF pulses.

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2 Release Summary

Software Component	MR-Bloch-Simulator
Version	https://gitlab.fme.lan/tactix/tactix-tutorials/-/tags/mr_simulator_v1.0.0 Hash: 4b63d78df610d57a36cb8b100de8d8681bf7a627
Embedding Product(s)	Not applicable
Purpose of this Release	Provision of the MR-Bloch-Simulator v1.0.0 software under AGPLv3-clause license for research and educational purposes. Not for clinical use.
OS and Compiler Details	OS: Windows 11 Node.js Version: 22.20.0
Embedded Libraries	The following libraries are used by the software. Detailed license information is contained in the delivered "third_party_licenses" folder and in the NOTICE file. • datgui (Apache License 2.0) • mathjs (Apache License 2.0) • parcel (MIT License) • statsjs (statsjs License) • threejs (MIT License)
Functionality	The first mode, the single-atom simulation, allows the application of user-configurable RF pulses to a single magnetization vector and the observation of the resulting evolution (precession, T1 and T2 relaxation). The second mode, the multi-atom simulation, simulates an ensemble of atoms to demonstrate slice selection. The simulation involves the application of a Z-gradient field and a user-configurable, frequency-selective RF pulse, leading to the selective excitation of a defined slice. A separate Demo Mode is also included. In this mode, the RF pulses are predefined and cannot be configured by the user.
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Safety Classification	None
Risk Measures	No risk measures defined.
Known Nonconformities and Restrictions	The software is not qualified for use as a medical product or as part thereof. No bugs or restrictions are known.